

DESCRIBING THE TYCHOS STRUCTURE – in reply to AI questions

The AI submitted a number of questions with regard to the TYCHOS model's framework:

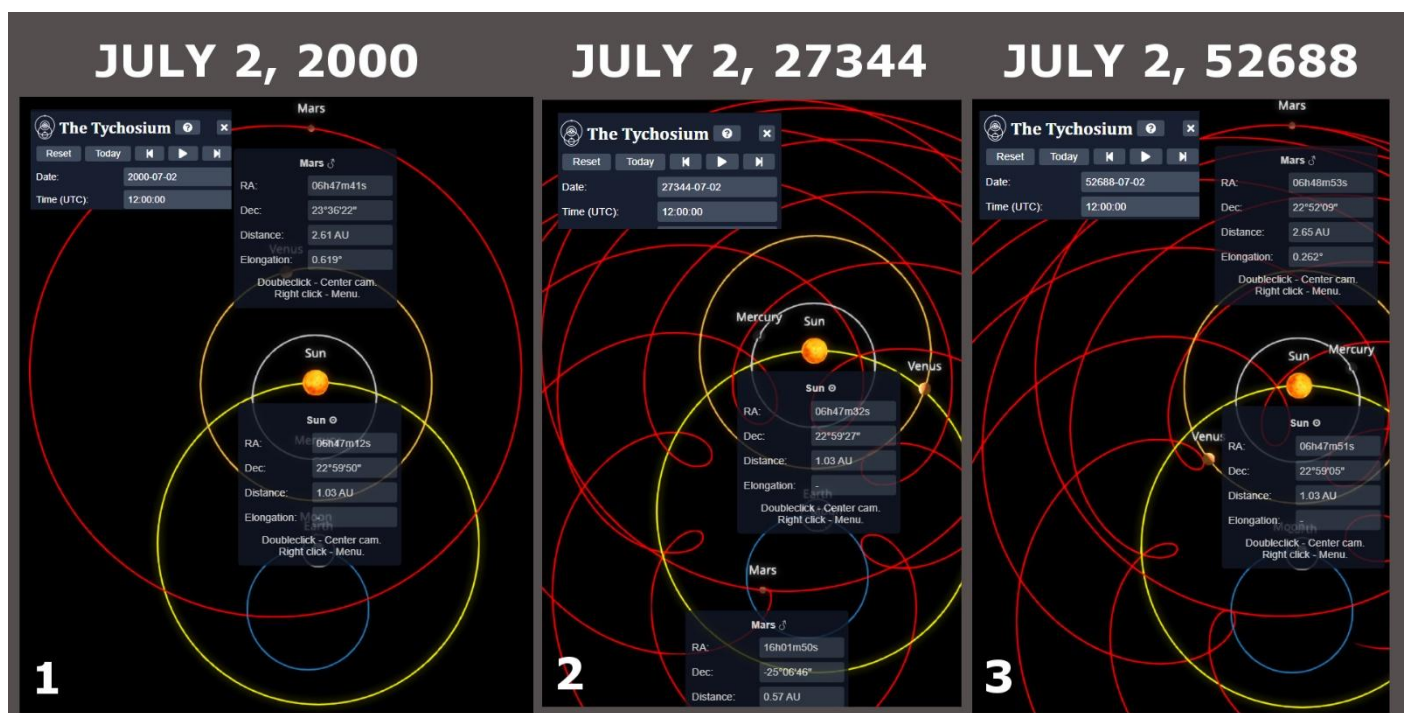
- 1 - What exactly does “binary companion” mean within the TYCHOS framework?
- 2 - Is Mars's orbit fundamentally defined relative to the Sun?
- 3 - Is there a physical common centre, or only an asymmetric primary-companion relationship?
- 4 - Which components are fundamental to Mars's orbit?
- 5 - What are the intended geometric roles of Mars deferent E and Mars deferent S?
- 6 - Should the annual carrier remain a primitive element, or emerge from another relationship?
- 7 - Which components should be adjustable and which should always be derived?
- 8 - Should Mercury and Venus literally share the solar-equatorial plane, or have separate planes close to it?

Rather than providing point-by-point answers to the AI's questions, my below response should nonetheless provide sufficient clarifications as to the general structure of my proposed TYCHOS Geoaaxial Binary System.

About the Sun-Mars binary relationship

In astronomy, it is a widely recognized fact that any given planetary cycle grows more accurate over time. For instance, Mars has three well-known approximate synodic cycles (of 47, 79 and 284 years), the precision of which increases progressively with each longer cycle.

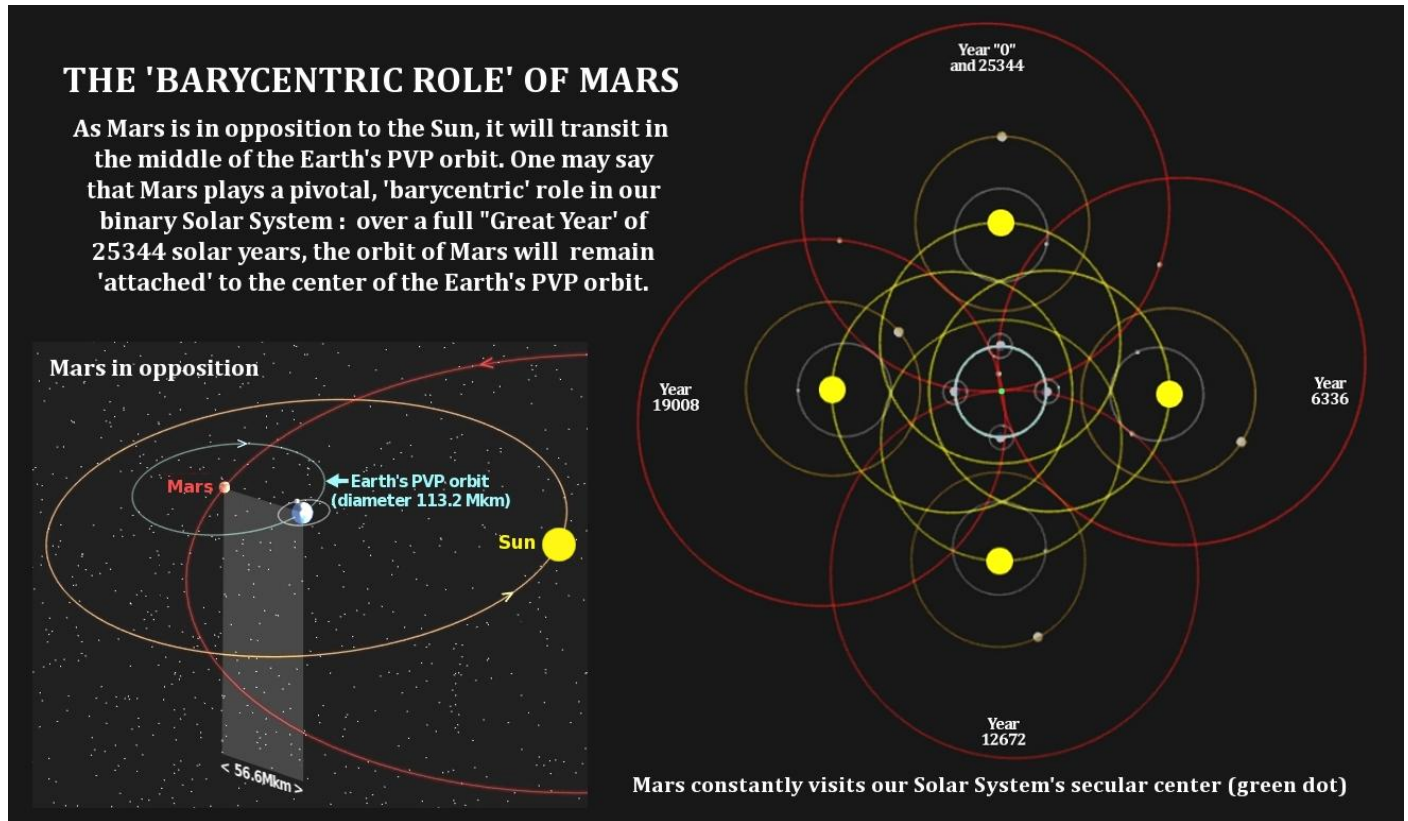
The TYCHOS Great Year is a cycle of 25344 years during which the Earth completes one of its PVP orbits. My below diagram shows how the Sun and Mars interact (in the Tychosium simulator) over two such Great Year cycles of 25344 years, thus demonstrating the perfect 2:1 binary resonance between the Sun and Mars.



1. July 2, 2000: Sun and Mars transit in near-exact conjunction at **≈RA 6h47 and ≈Dec 23°**
2. July 2, 27344: Sun returns to **≈RA 6h47 and ≈Dec 23°** while Mars transits in opposition at only 0.57 AU from Earth (a distance which closely matches the radius of the PVP orbit).
3. July 2, 52688: Sun and Mars re-conjunct at **≈RA 6h47 and ≈Dec 23°**, Mars having completed exactly 2 periods of 25344 years, 50688 years being the most precise long cycle of Mars.

About the (secular) center of the TYCHOS binary system

Having established that the Sun and Mars exhibit a perfect 2:1 ratio over a period of 2×25344 years, I then realized that the rim of the Martian orbit remains effectively 'attached' to the center of the PVP orbit at all times. My below graphic illustrates how this plays out over a full TYCHOS Great Year (divided in four parts).



One may thus reasonably infer that the 'secular barycenter' of the entire system is located in the middle of the Earth's PVP orbit. In any event, this is what the TYCHOS geometric framework clearly suggests – and how the Tychosium simulator empirically plots the system's motions over 25344 years.

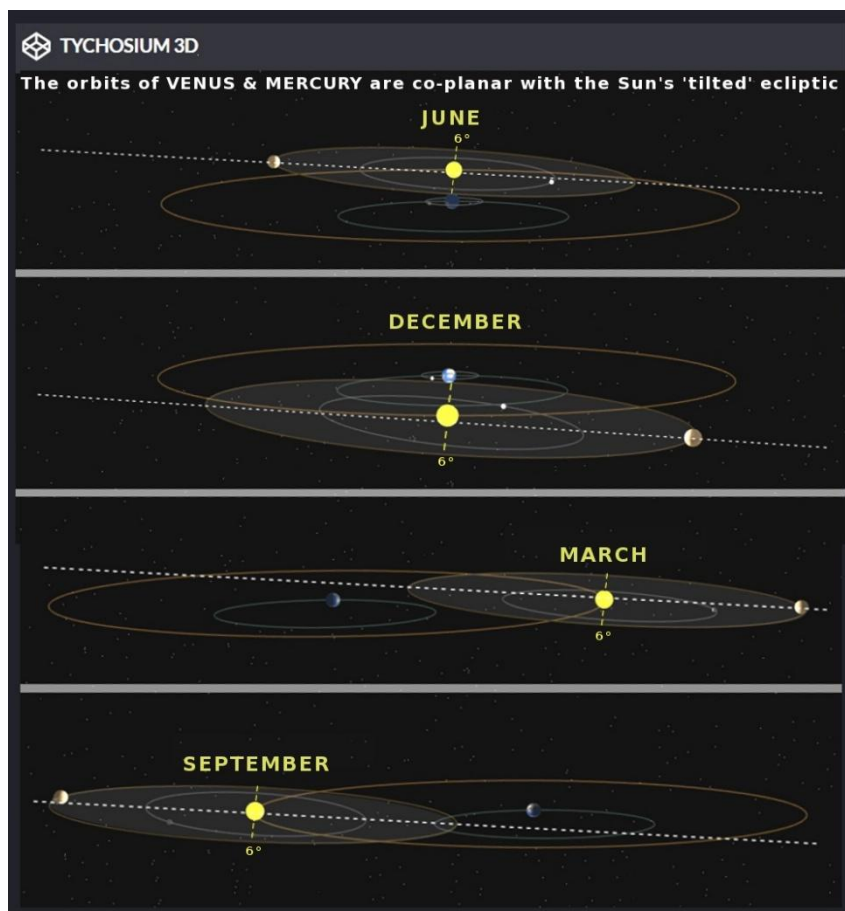
Whether this allows us to define the center of the PVP orbit as our system's "secular barycenter" is up for debate, but this would likely be the conclusion of any kinematic laboratory experiment performed with finely calibrated magnets revolving around a central 'magnetic field'. To be sure, Isaac Newton's 'universal gravitation' is but a mathematical construct which has now been roundly falsified by modern observations of distant galaxies.

About the actual orbital speeds of the bodies in our Solar System

A recent discovery of the TYCHOS research should also be of interest to the AI. Namely, that the Tychosium's new "show speeds" function shows that almost all the bodies in our Solar System (except Venus and Mercury - but including Halley and Eros) actually travel at roughly the same speed as the Sun (≈ 30 km/s). The reason for this is methodically expounded and illustrated in this forum post of mine: <https://forum.tychos.space/t/the-end-of-dark-matter-and-of-kepler-s-laws/751>

About Venus and Mercury and their orbits' axial parallelism (coplanar with the Sun's ecliptic)

The Tychosium simulator also reveals how the orbits of Venus and Mercury are coplanar with the Sun's equatorial plane (which is tilted at ≈ 6 or 7°) – and remain so at all times, in what is known as 'axial parallelism'.



Remarkably, Venus and Mercury can also be shown – in various ways - to be 'locked' at a 2:1 ratio, much like the Sun and Mars (in the TYCHOS of course, the moonless Venus and Mercury are the two moons of the Sun).

- Venus returns to the same geocentric ephemerides exactly every 2000 years
- Mercury returns to the same geocentric ephemerides exactly every 1000 years
- Venus' mean / average retrograde period is 45.6 days
- Mercury's mean / average retrograde period is 22.8 days

As I mentioned in my 2023 TYCHOS book (yet failed to clarify), the rotational speeds of Mercury and Venus around their axes (vis-à-vis absolute space) could also theoretically be said to be locked in a 2:1 ratio (about 5.4km/h vs 2.7 km/h). But as I later realized, it can be more correctly said that "tidally-locked" moons (including our own Moon) **do not rotate at all** around their axes. To understand this, one only needs to mentally visualize moons as magnets attached to their hosts with a yo-yo string. Yes, they do rotate by 360° in absolute space during each of their revolutions, but this rotation is not intrinsic / proprietary to the moons themselves.

I will candidly admit that I don't fully grasp the AI's queries 4,5,6, and 7 – but hopefully this short PDF of mine will at least clarify some key aspects of the TYCHOS framework and its core geometric configuration.